News Credibility Detection & Writing Style Analysis System

Project Report — Milestone 2

Agentic AI System: ML + RAG + LLM Integration

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Technologies	Python, Scikit-learn, TF-IDF, Logistic Regression, Groq API (LLaMA 3.1), News API, RAG, Streamlit
Deployment	Streamlit Cloud · GitHub Repository

"This project showcases how modern AI systems move beyond prediction to ***Explain, Validate, and Reason*** *— not just classify."*

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1. Introduction

The rapid spread of misinformation in digital media demands automated systems that can evaluate news credibility in an interpretable and reliable manner. Most existing fake news detection models focus only on binary classification and behave as black-box predictors, offering little explanation of *why* an article is considered fake or real.

This project addresses this gap by developing a **hybrid news credibility analysis system** that combines statistical text classification with behavioural writing-style analysis.

Project Vision & Milestones

The complete project is designed in two progressive stages:

- **Milestone 1 (Completed & Submitted):** Machine learning–based fake news classification with interpretable writing-style metrics.
- **Milestone 2 (Current Report):** Extended credibility analysis framework integrating RAG-based evidence retrieval and LLM-powered reasoning for a full Agentic AI pipeline.

2. Milestone 1 — Summary of Completed Work

Milestone 1 established the foundational machine learning pipeline for fake news detection. The core components built in this milestone are briefly summarized below.

2.1 Core ML Pipeline

- **Dataset:** ISOT Fake News Dataset (`True.csv` + `Fake.csv`) from Kaggle, with stratified 80/20 train-test split.
- **Text Preprocessing:** Lowercasing, removal of publisher identifiers (e.g., "(Reuters)"), punctuation/number removal, leakage keyword elimination, and whitespace normalization.
- **Feature Engineering:** TF-IDF Vectorization (max 20,000 features, unigrams + bigrams, min_df=5, max_df=0.9).
- **Model:** Logistic Regression classifier producing a probability-based credibility score scaled to 0–100.

2.2 Writing Style Analysis Module

Alongside ML classification, Milestone 1 introduced a rule-based behavioural analysis module extracting the following linguistic credibility signals:

- **Sensational Language Score** — detects dramatic emotional words.
- **Clickbait Framing Score** — identifies viral framing patterns.
- **Exaggeration Score** — flags extreme wording and hyperbole.
- **Visual Emphasis Score** — detects excessive CAPS, !, and ?.
- **Emotional Intensity** — VADER-based sentiment magnitude analysis.

2.3 Key Achievements in Milestone 1

- Strong baseline model with reliable fake news classification performance.
- Leakage prevention through systematic preprocessing.
- Interpretable credibility scoring (0–100%) instead of binary output.
- Structured analytical report combining ML prediction and style metrics.

3. Milestone 2 — Objectives

Building on the Milestone 1 foundation, Milestone 2 aimed to transform the basic ML classifier into a complete, production-ready Agentic AI system. The specific objectives were:

1. Detect whether a news claim is **REAL** or **FAKE** using the trained ML model from Milestone 1.
2. Retrieve relevant real-world news evidence in real time using a **Retrieval-Augmented Generation (RAG)** approach with News API integration.
3. Provide logical, evidence-grounded explanations using a **Large Language Model (LLM)** via the Groq API.
4. Handle edge cases including absence of retrieved news, misleading queries, and ambiguous claims.
5. Ensure system security — preventing data leakage, API key exposure, and prompt injection attacks.
6. Build a deployable, production-like web interface using **Streamlit**.

4. System Architecture

The Milestone 2 system follows an end-to-end **Agentic AI pipeline** where each component plays a clearly defined role.

4.1 System Workflow

End-to-End Pipeline

1. **User Input** — News text entered via Streamlit UI
   ↓
2. **ML Model** (`predict_news`) — Classify as REAL/FAKE + credibility score + style metrics
   ↓
3. **RAG System** (`fetch_news`) — Retrieve relevant articles via News API
   ↓
4. **LLM Analyzer** (`analyze`) — Groq API / LLaMA 3.1 reasoning over claim + evidence
   ↓
5. **Fallback Reasoning** — If no news retrieved, LLM reasons from general knowledge
   ↓
6. **Confidence Score** — Combined ML + evidence-based confidence computation
   ↓
7. **Streamlit UI** — Structured results displayed to the user

5. System Components — Detailed Description

5.1 Machine Learning Model (Retained from Milestone 1)

The ML model from Milestone 1 was retained and integrated as the first-stage classifier in the Milestone 2 pipeline.

- **Function:** `predict_news(text)`
- **Model:** Logistic Regression with TF-IDF feature representation

- **Output:** Binary classification (REAL/FAKE) + credibility score (%) + linguistic style metrics

The five style metrics extracted (Sensationalism, Clickbait, Exaggeration, Visual Emphasis, Emotional Intensity) continue to enhance explainability and serve as supplementary signals for the LLM reasoning stage.

5.2 RAG System — News Retrieval Module

The Retrieval-Augmented Generation (RAG) component was newly built in Milestone 2 to fetch live, real-world news evidence relevant to the input claim.

- **API Used:** News API
- **File:** `rag/news_fetcher.py`

5.2.1 Key Improvements Implemented

- **Query Cleaning:** Natural language input is converted into clean, keyword-based search queries for effective retrieval.
- **Dynamic Retrieval:** Removed hardcoded restrictive filters (e.g., the earlier `"covid"` topic filter) to allow flexible, topic-agnostic searches.
- **Fallback Query:** If the initial search returns no results, the system automatically retries with an expanded query (`"news + [query]"`).
- **Structured Output:** Each retrieved article is returned in a standardized JSON format containing `title`, `content`, `url`, and `source`.

5.3 LLM Analyzer — Groq API Integration

The LLM Analyzer constitutes the reasoning core of the Milestone 2 system. It receives the user's claim alongside retrieved news evidence and produces an evidence-grounded explanation.

- **API Used:** Groq API
- **Model:** `llama-3.1-8b-instant`
- **Function:** `analyze(query, docs)`

5.3.1 LLM Responsibilities

- Analyze the input claim in the context of retrieved evidence.
- Determine whether the claim is **Supported**, **Contradicted**, or **Unclear** based on evidence.
- Generate a structured, logical explanation of the verdict.

5.3.2 Safety & Security Features

- Rejects attempts to extract API keys or system prompt content.
- Ignores prompt injection attempts embedded within user queries.
- Declines requests involving personal data queries.
- Balanced prompt engineering ensures normal analytical reasoning without over-refusal.

5.4 Fallback Reasoning

A critical reliability improvement in Milestone 2 is the **Fallback Reasoning** mechanism, which handles cases where the News API returns no relevant articles.

- **Problem Addressed:** Without fallback logic, the system previously failed or returned unhelpful responses when no news was available.
- **Solution:** When no documents are retrieved, the LLM is invoked with a `fallback_docs` flag and instructed to reason from general knowledge while clearly communicating uncertainty.
- **Behavior:** Avoids hallucination, explains evidence availability, and provides reasoned assessment.

Fallback Example

Input: "Aliens landed in Delhi yesterday"

Result: No news evidence found → LLM applies general knowledge reasoning → Claim assessed as likely **FAKE** with explanation of absence of corroborating evidence.

5.5 Confidence Score Computation

Milestone 2 introduces a composite **Confidence Score** that combines the ML model's certainty with evidence availability.

Confidence Score Formula

$$\text{Confidence (with evidence)} = (\text{ML Score} \times 0.6) + 20$$

$$\text{Confidence (without evidence)} = \text{ML Score} \times 0.7$$

Purpose: Combines ML certainty + evidence availability into a single interpretable percentage.

5.6 Streamlit Web Interface

A fully functional web interface was developed and deployed using Streamlit.

5.6.1 UI Features

- Input text box for entering news claims.
- ML verdict display with credibility percentage.
- Confidence score visualized as a progress bar.
- Writing style metrics displayed in a clean JSON view.
- Retrieved news articles with title, content preview, clickable source link, and publisher name.
- LLM-generated explanation in a dedicated section.

5.6.2 UI Design Improvements

- Two-column layout for organized information presentation.
- Clear section separation for distinct result categories.
- Comprehensive error handling for API failures and edge cases.

6. Security & Safety Implementation

Milestone 2 places significant emphasis on real-world security considerations, reflecting a production-level engineering approach.

Protection	Implementation Details
API Key Security	All API keys stored as environment variables (`.env`). Never hard-coded in source files.
Prompt Injection Defense	LLM system prompt designed to detect and ignore malicious instructions embedded in user inputs.
Data Privacy	No personal data is exposed or logged. System prompt contents are never leaked to users.
Safe Topic Handling	Sensitive queries receive neutral, factual responses. No harmful instructions are ever generated.

7. Testing Strategy & Results

7.1 Test Categories

The system was tested across four distinct categories:

- **Fake News Tests:** COVID misinformation claims, conspiracy theories.
- **Real News Tests:** GDP growth reports, government policy updates.
- **Edge Case Tests:** Queries with no available news, very short inputs.
- **Security Tests:** API key extraction attempts, prompt injection attacks.

7.2 Sample Test Results

Input Claim	System Output
"Aliens landed in Delhi yesterday"	**FAKE** — No evidence found; fallback reasoning applied.
"India GDP growth rate increased"	**REAL** — Corroborating articles retrieved; LLM confirms.
"Vaccine causes autism"	**FAKE** — Scientific reasoning applied; contradicting evidence cited.
API key extraction attempt	Safely **rejected** — LLM responds with neutral disclaimer.
Short query: "war"	**Edge case handled** — Partial analysis with uncertainty acknowledged.

8. Challenges & Solutions

#	Challenge	Solution Implemented
1	No news articles retrieved for certain queries	Query cleaning + automatic fallback retry with expanded query.
2	Irrelevant articles returned by News API	Removed restrictive hardcoded filters for broader, flexible searches.
3	LLM over-refusing legitimate queries	Refined system prompt to balance safety with normal analytical reasoning.
4	Groq API model deprecation mid-project	Updated model identifier to `llama-3.1-8b-instant`.
5	API key security in deployment environment	Migrated to environment variables and Streamlit Secrets.

9. Deployment

Attribute	Details
Platform	Streamlit Cloud
Entry Point	`Milestone2/app.py`
Repository	Push to GitHub; deploy via Streamlit Cloud dashboard
API Key Management	Environment variables / Streamlit Secrets (never hardcoded)
Live Access	Accessible via public Streamlit URL after deployment

10. Key Learnings

- **RAG requires query engineering** — Natural language inputs must be converted to clean keywords for effective retrieval.
- **LLMs must be controlled via prompts** — Careful prompt design balances helpfulness with safety, avoiding both over-refusal and harmful outputs.
- **Systems must handle absence of data** — Fallback mechanisms are essential for

real-world reliability.

- **Security is critical in production AI** — Prompt injection, API exposure, and data privacy must be explicitly addressed from the start.
- **UI presentation affects usability** — Well-structured interfaces improve trust and interpretability of AI-generated outputs.

11. Milestone 1 vs. Milestone 2 — Comparison

Feature	Milestone 1	Milestone 2
Classification	ML only	ML + RAG evidence + LLM reasoning
Explainability	Style metrics only	Style metrics + LLM explanations
Evidence	None	Real-time retrieval via News API
Edge Case Handling	Not addressed	Fallback reasoning mechanism
Security	Basic	Prompt injection defense + env variable management
User Interface	None	Full Streamlit web application
Deployment	Google Colab notebook	Streamlit Cloud production deployment
System Type	Classifier	Agentic AI (multi-step reasoning pipeline)

12. Conclusion

Milestone 2 successfully transformed the foundational fake news classifier from Milestone 1 into a complete, production-ready **Agentic AI system**. By integrating Machine Learning, Retrieval-Augmented Generation, and Large Language Model reasoning into a unified pipeline, the system now delivers multi-step, evidence-grounded fact-checking rather than simple binary classification.

The system addresses real-world requirements including dynamic evidence retrieval, transparent reasoning, robust fallback handling, API security, and a deployable web interface. These advancements reflect a principled approach to modern AI system design.

The project demonstrates how contemporary AI systems can move beyond prediction toward **explanation, validation, and reasoning** — combining the reliability of machine learning with the contextual intelligence of large language models to produce a transparent, robust, and practically deployable misinformation detection tool.

> *"This project showcases how modern AI systems move beyond prediction to*
> ***Explain, Validate, and Reason** — not just classify."*

References & Resources

- **Dataset:** ISOT Fake News Dataset — https://www.kaggle.com/datasets/csmalarkodi/isot-fake-news-dataset
- **GitHub Repository:** https://github.com/GEN-NOAA/FakeNewsDetector
- **Milestone 1 Deployment:** https://fakenewsdetectorgit-hhwwpb9eqfzjj2uywqqmrw.streamlit.app/
- **Milestone 1 Notebook:** https://colab.research.google.com/drive/1rTZb2ngZULBXQHOdU53e
- **Groq API:** https://groq.com Model: `llama-3.1-8b-instant`
- **News API:** https://newsapi.org